

🕶 Weekly Eye Trends Dashboard

Automated Google Trends report for eyewear & eye health, delivered every Monday via email and logged to Google Sheets.

Project Structure

```
your-repo/
├── .github/
│   └── workflows/
│       └── weekly_trends.yml    ← GitHub Actions schedule
├── trends_dashboard.py          ← Main script
├── requirements.txt
└── README.md
```

One-Time Setup

Step 1 — SerpAPI Key

1. Sign up at serpapi.com
 2. Copy your API key from the Dashboard
 3. Free tier = 100 searches/month (covers ~12 keywords/week comfortably)
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Step 2 — Gmail App Password

1. Enable **2-Factor Authentication** on your Google account
 2. Go to **Google Account → Security → App Passwords**
 3. Generate a password for "Mail / Other"
 4. Save the 16-character password — this is your `EMAIL_PASSWORD`
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Step 3 — Google Sheets + Service Account

This lets the script write to your Sheet without prompting for login.

1. Go to console.cloud.google.com
 2. Create a new project (or use an existing one)
 3. Enable these two APIs:
 - **Google Sheets API**
 - **Google Drive API**
 4. Go to **IAM & Admin → Service Accounts → Create Service Account**
 - Name it anything (e.g. `trends-bot`)
 - Skip optional steps, click Done
 5. Click the service account → **Keys → Add Key → JSON**
 - Download the JSON file — you'll need the full contents
 6. Create a new Google Sheet at sheets.google.com
 - Copy the Sheet ID from the URL:
`https://docs.google.com/spreadsheets/d/THIS_IS_THE_ID/edit`
 7. Share the Sheet with the service account email (looks like `trends-bot@your-project.iam.gserviceaccount.com`) — give it **Editor** access
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Step 4 — GitHub Repository Secrets

Push your files to a GitHub repo, then go to: **Settings → Secrets and variables → Actions → New repository secret**

Add all 6 secrets:

Secret Name	Value
<code>SERPAPI_KEY</code>	Your SerpAPI key
<code>EMAIL_FROM</code>	Your Gmail address
<code>EMAIL_TO</code>	Recipient email (can be the same)

EMAIL_PASSWORD	Your 16-char Gmail App Password
GSHEET_ID	The Sheet ID from the URL
GOOGLE_CREDS_JSON	The entire contents of the service account JSON file

⚠ For `GOOGLE_CREDS_JSON`, paste the raw JSON (including the outer `{ }` braces) as a single secret value.

Step 5 — Push & Test

```
git add .
git commit -m "Add weekly eye trends dashboard"
git push
```

Then go to **Actions → Weekly Eye Trends Dashboard → Run workflow** to trigger it manually and confirm everything works before waiting for Monday.

Google Sheets Output

The script auto-creates a **“Weekly Log”** sheet tab with these columns:

Week	Keyword	Avg Interest (0–100)	Peak	Direction
April 28, 2025	blue light glasses	74.3	100	
April 28, 2025	dry eyes	61.2	88	
...	...	...	...	...

Each Monday appends new rows — building a historical log you can chart over time.

Customizing Keywords

Edit the `KEYWORDS` list in `trends_dashboard.py`:

```
KEYWORDS = [
    "eye exam", "dry eyes", "eye drops", "vision correction",
```

```
"blue light glasses", "prescription sunglasses", "contact lenses",  
"oversized glasses", "cat eye glasses", "transition lenses",  
]
```

- Max 5 per SerpAPI call (Google Trends limit), but you can have as many groups as you like
 - Add competitor brands: "Warby Parker", "Zenni Optical", "1-800 Contacts"
 - Add seasonal terms: "UV protection glasses", "ski goggles"
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Adjusting the Schedule

Edit the cron expression in `weekly_trends.yml`:

```
- cron: "0 12 * * 1"    # Every Monday at 12:00 UTC = 8:00 AM Eastern
```

Cron format: minute hour day month weekday

Use crontab.guru to build custom schedules.